A3-PS033-TC318

DNS-Based Dicovery of Diagnostic Services

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Problem Description

TrustDiag addresses weaknesses in discovering and invoking diagnostic services.

- Hardcoded IP addresses and URLs become stale when endpoints move.
- DNS data and service metadata may be untrusted or malformed.
- DNSSEC alone does not verify the API server TLS identity or authorize a client.
- Unrestricted diagnostic targets and inconsistent errors weaken safe operation.

The goal is verifiable discovery followed by authenticated, authorized, and auditable execution.

Background

DNS provides naming and records that can publish service connection information.

SVCB and HTTPS records can advertise connection parameters and alternative endpoints [2]. DNS-AID describes a DNS-based discovery pattern for AI agents, but remains an Internet-Draft; TrustDiag adapts relevant concepts to diagnostic services rather than claiming that diagnostics are standardized by that draft [1]. DNSSEC provides origin authentication and integrity for DNS data, but not confidentiality, API permissions, or proof of the HTTPS server identity [3]. TLS protects the client-server connection, while the API independently authenticates the client and authorizes each operation [4], [5]. Endpoint changes still depend on authorized DNS updates and cache/TTL expiry.

Literature & RFC Review

The most relevant proposals and standards cover complementary parts of the problem.

DNS-AID [1] describes discovery of AI agents using existing DNS mechanisms; it is a working Internet-Draft, not an approved RFC. RFC 9460 [2] specifies SVCB and HTTPS records for service connection information. RFC 4033 [3] describes DNSSEC data-origin authentication and integrity, which do not replace TLS verification or API authorization. TLS 1.3 [4] protects client-server communications. RFC 6750 [5] describes bearer-token use for protected HTTP resources and stresses protection of tokens in storage and transport. TrustDiag combines these independent controls in a diagnostic discovery workflow and evaluates their interaction through negative security tests.

Proposed Solution

TrustDiag combines DNS discovery with layered security checks.
Services are published in a controlled DNS zone. The client discovers the endpoint and validates DNSSEC status, service identity, capability, protocol, and target policy. Untrusted results are rejected when validation is required. The client then verifies the server certificate and hostname over HTTPS. The API checks a bearer token, authorizes each operation, and restricts targets to an allowlist. It returns structured results and safe audit logs. TrustDiag supports latency, connectivity, and routing; simulated output is labelled.

Optimization Proposed

Planned design-level comparison; each improvement must be verified by testing.

Control	Static baseline	TrustDiag target
Discovery	Fixed endpoint URLs	Named endpoints discovered via DNS
DNS integrity	DNS answer accepted as-is	Require DNSSEC when policy demands
Metadata	Service details assumed	Validate identity, capability, protocol
Server identity	Endpoint configured manually	Verify HTTPS certificate and hostname
API access	Limited operation-level control	Bearer token plus server authorization
Diagnostics	Loose target controls / logs	Allowlist targets; audit results safely

Solution Architecture & Design

TrustDiag separates discovery validation, TLS identity, and API authorization.

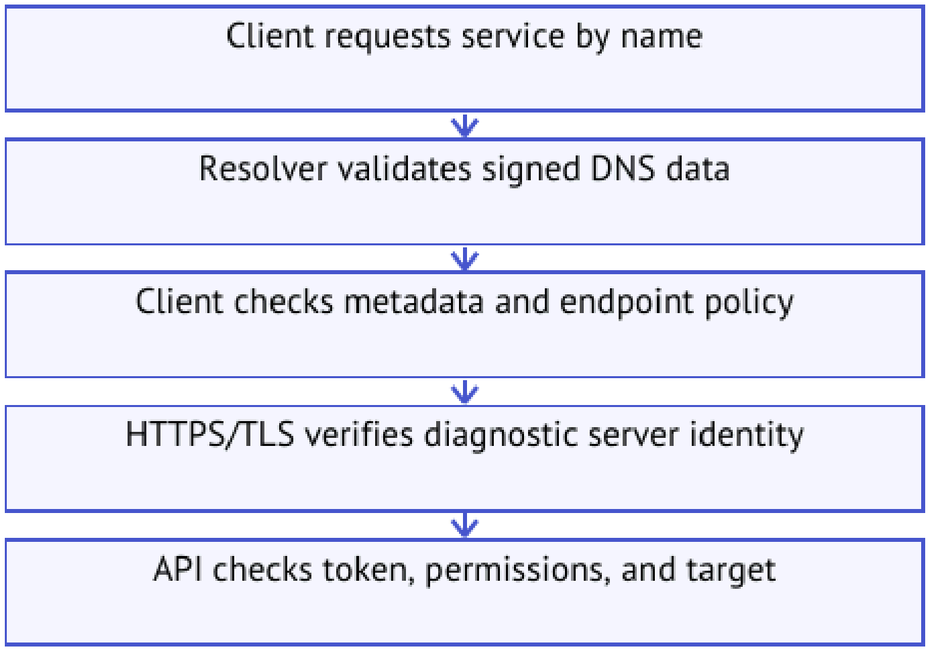


Figure: TrustDiag System Overview

Data Flow Diagram

DNS validation gates discovery; HTTPS requests reach only authorized APIs. Structured diagnostic results and security events are recorded.

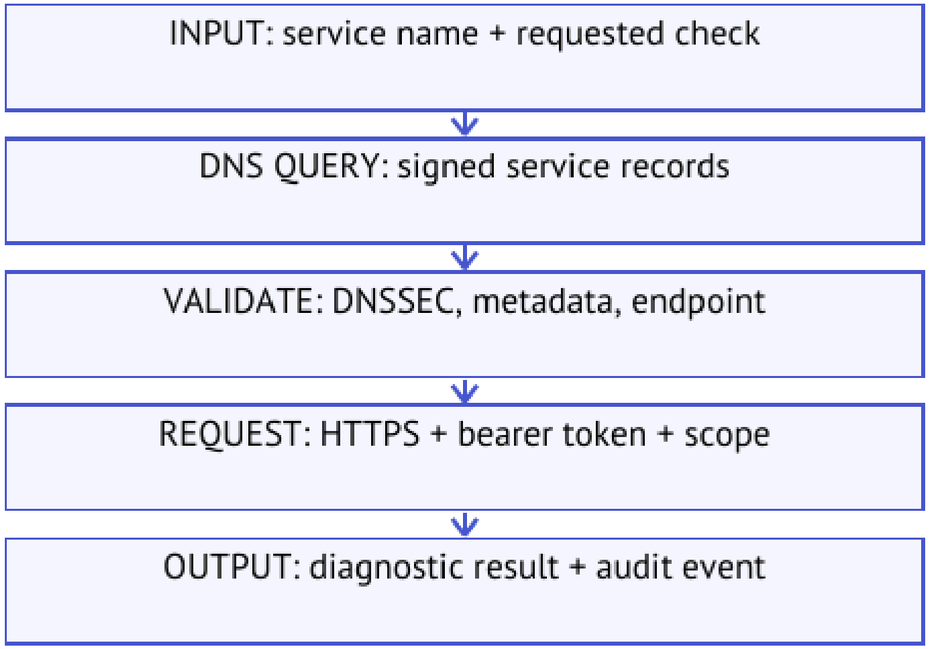


Figure: TrustDiag Data Flow Diagram

Deployment Timeline

Deployment milestones are staged to validate each trust boundary before integration.

Phase 1 - confirm scope, DNS-AID revision, metadata schema, and target policy.

Phase 2 - configure the signed DNS zone and validating resolver; publish service records.

Phase 3 - implement latency, connectivity, and routing APIs with HTTPS and access controls.

Phase 4 - integrate the client and server; run functional and adversarial tests.

Phase 5 - compare baseline with TrustDiag, record measurements, and prepare the report and demo.

Final review confirms that mandatory security failures stop protected invocation.

Bibliography I

[1] J. Mozley et al., “DNS for AI Discovery,” IETF Internet-Draft, draft-mozleywilliams-dnsop-dnsaid-02, May 2026. Active working draft; not an approved RFC.

[2] B. Schwartz, M. Bishop, and E. Nygren, “Service Binding and Parameter Specification via the DNS (SVCB and HTTPS Resource Records),” RFC 9460, Nov. 2023.

[3] R. Arends et al., “DNS Security Introduction and Requirements,” RFC 4033, Mar. 2005.

[4] E. Rescorla, “The Transport Layer Security (TLS) Protocol Version 1.3,” RFC 8446, Aug. 2018.

[5] M. Jones and D. Hardt, “The OAuth 2.0 Authorization Framework: Bearer Token Usage,” RFC 6750, Oct. 2012.